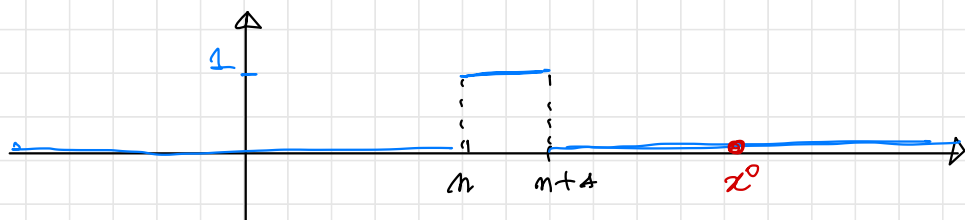


Controesempio 1)

$\exists f_n \rightarrow 0$ per q.o. $x \in \mathbb{R}$ ma $\int_{\mathbb{R}} |f_n| \not\rightarrow 0$
(ovvero $f_n \not\rightarrow 0$ in $L^1(\mathbb{R})$)

$$f_n(x) = \chi_{(n, n+1)}(x) = \begin{cases} 1 & x \in (n, n+1) \\ 0 & x \notin (n, n+1) \end{cases}$$



• $f_n(x) \rightarrow 0$ q.o. $x \in \mathbb{R}$

fisso $x^0 \in \mathbb{R}$ e considero $\{f_n(x^0)\}$,

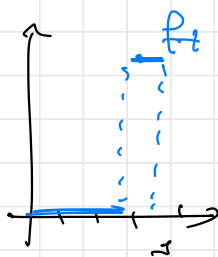
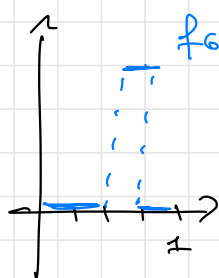
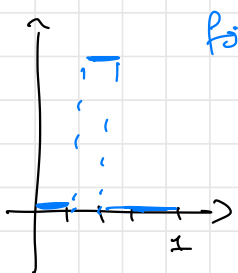
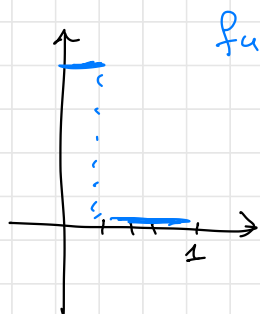
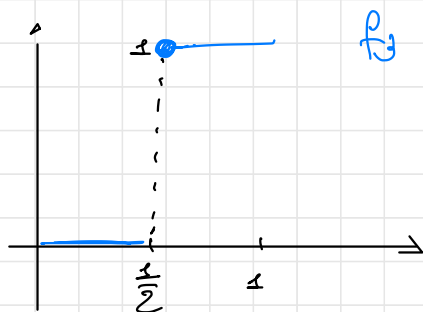
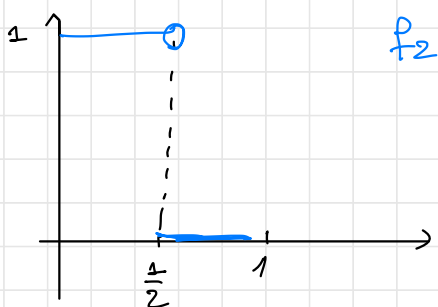
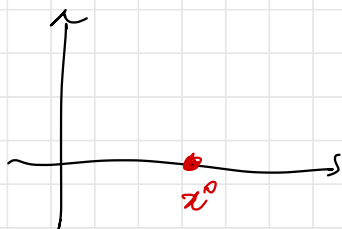
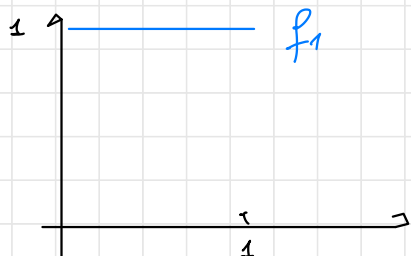
per n abbastanza grande, $f_n(x^0) \equiv 0$

$$\bullet \|f_n\|_1 = \int_{\mathbb{R}} |f_n| = \int_{\mathbb{R}} f_n = 1$$

Proposizione $\Rightarrow f_n$ non ha limite in $L^1(\mathbb{R})$

Controesempio 2)

$\exists \{f_n\} \rightarrow 0$ in $L^1(0,1)$ ma $f_n(x) \not\rightarrow 0$ q.o. su $(0,1)$.



$\bullet \quad \|f_n\|_{L^1} \rightarrow 0$

$\bullet \quad \forall x^0 \in (0,1), \quad f_n(x^0) \not\rightarrow 0$

$\exists \underbrace{f_{k(n)}}_{\substack{\text{---} \\ \text{---} \\ 1}}(x^0) \not\rightarrow 0$

Proposizione

Se $f_n \rightarrow 0$ in $L^1(E) \Rightarrow \exists |f_{k(n)}| \rightarrow 0$
q.o. su E .

Conseguenze:

Se una successione converge in $L^1(E)$,
e converge q.o. a f ,
allora il limite in $L^1(E)$ è uguale a f .

• $f_n \rightarrow f$ q.o. su E $\overset{\text{Proprietà del limite}}{\Rightarrow}$

$\forall k(n) \quad f_{k(n)} \rightarrow f$ q.o. su E

• $f_n \rightarrow g$ in $L^1(E)$ $\overset{\text{PROP}}{\Rightarrow}$

$\exists k(n): f_{k(n)} \rightarrow g$ q.o. su E

$f_{k(n)} \xrightarrow{\quad} f \text{ q.o. su } E$
 $\searrow \quad \quad \quad g \text{ q.o. su } E$

$\Downarrow$ unicità del lm

$f = g$ q.o. su E .

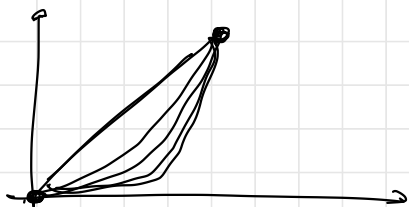
Recall: Riemann:

$$f_n \xrightarrow[\text{continue uniformemente}]{} 0 \text{ su } [a, b] \implies \lim \int_a^b f_n = 0$$

$$\max_{[a, b]} |f_n| \rightarrow 0$$

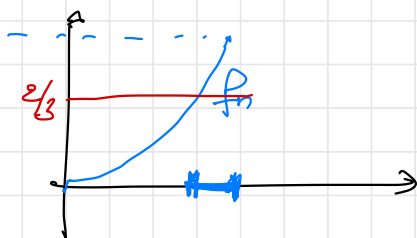
Esempio $[0, 1] \quad f_n(x) = x^n \rightarrow 0 \quad \forall x \in [0, 1)$

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1} \rightarrow 0$$



$$\max_{x \in [0, 1]} |f_n(x)| = 1$$

$$\parallel f_n \parallel_\infty$$



$$|f_n| \leq \frac{2}{3}$$

Teorema di conv. dominata (Lebesgue)

Sia $f_n \in L^1(E)$ e sia $f_n \rightarrow f$ q.o. su E .

Supponiamo $\exists g \in L^1(E)$ indipendente da n tale che

$$|f_n(x)| \leq g(x) \quad \text{q.o. su } E. \quad \underbrace{\forall n \in \mathbb{N}}_{\text{(definitivamente)}}$$

Allora $f_n \rightarrow f$ in $L^1(E)$.

Esempi

1) $f_n(x) = x^n \quad x \in (0,1]$

$$|f_n(x)| \leq 1 \quad g(x) = 1 \text{ dominante}$$

2) $f_n(x) = \chi_{(n, n+1)}$

$$\exists g \in L^1(\mathbb{R}) \text{ tale che } |f_n(x)| \leq g(x) \text{ q.o. su } \mathbb{R} \\ \text{"} \quad \quad \quad \forall n \geq 1 \\ \chi_{(a, +\infty)}$$

Teorema di conv. monotona (Beppo Levi)

Sia $\{f_n\} \in L^1(E)$.

Supponiamo che

$$f_n \geq 0 \text{ q.o. su } E, \quad f_{n+1} \geq f_n \text{ q.o. su } E$$

Allora:

$$\int_E \lim_n f_n = \lim_n \int_E f_n$$

$$\bullet \quad g_n \leq 0 \text{ q.o. su } E \quad g_{n+1} \leq g_n \text{ q.o. su } E$$

$$f_n = -g_n$$

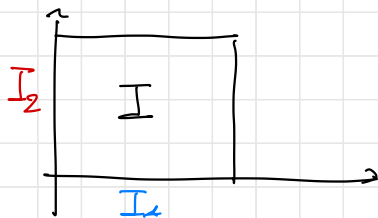
$$\bullet \quad f_n \in L^1(E), \quad \boxed{f_n \geq 0}$$

$$\int_E \sum_n f_n = \sum_n \int_E f_n$$

$$\left(S_N \right) = \sum_{n=0}^N f_n \geq 0 \quad S_{N+1} \geq S_N$$

Integrali multipli

$$I \subseteq \mathbb{R}^{m+n} \quad I = I_1 \times I_2 \subseteq \mathbb{R}^m \subseteq \mathbb{R}^n$$



$$x \in I \iff x_1 \in I_1, x_2 \in I_2$$

(x₁, x₂)

Teorema di Fubini

Sia f $\mathbb{L}$ -integrabile (in $\mathbb{R}^2$) in I .
in $\mathbb{R}^{m+n}$

Allora:

1) Per q.o. $x_1 \in I_1$, $x_2 \mapsto f(x_1, x_2)$ $\mathbb{L}$ -integrabile su I_2

2) $x_1 \mapsto \int_{I_2} f(x_1, x_2) dx_2$ $\mathbb{L}$ -integrabile su I_1

$$\begin{aligned} 3) \quad \int_I f(x_1, x_2) dx_1 dx_2 &= \int_{I_1} \left[\int_{I_2} f(x_1, x_2) dx_2 \right] dx_1 \\ &= \int_{I_2} \left[\int_{I_1} f(x_1, x_2) dx_1 \right] dx_2 \end{aligned}$$

Teorema di Tonelli

Sia $f \geq 0$ misurabile su $I = I_1 \times I_2$.

Supponiamo che:

- 1) Per q.o. $x_1 \in I_1$, $x_2 \mapsto f(x_1, x_2)$ $\mathcal{L}$ -integrabile su I_2
- 2) $x_1 \mapsto \int_{I_2} f(x_1, x_2) dx_2$ $\mathcal{L}$ -integrabile su I_1

Allora,

f è $\mathcal{L}$ -integrabile su $I = I_1 \times I_2$

$$\left(\begin{array}{c} \Rightarrow \\ \text{Fubini} \end{array} \right) \int_I f = \int_{I_1} \left(\int_{I_2} f dx_2 \right) dx_1 = \int_{I_2} \left(\int_{I_1} f dx_1 \right) dx_2$$

Obs. Se f cambia segno e non so apriori
che f $\mathcal{L}$ -integrabile su I ,

posso considerare $|f|$ e provare ad applicare

Tonelli: Se $|f|$ soddisfa 1) e 2), allora

$|f|$ $\mathcal{L}$ -integrabile su $I = I_1 \times I_2 \Rightarrow$

f $\mathcal{L}$ -integrabile su $I = I_1 \times I_2 \Rightarrow$ per Fubini
 $\rightarrow$ "posso integrare per fetto"

Spazi L^p e spazi di Lebesgue

Def. Sia $p \in [1, +\infty)$, E misurabile in $\mathbb{R}^n$

$$L^p(E) := \{ f: E \rightarrow \mathbb{R} \text{ tale che } \int_E |f|^p < +\infty \}$$

↑ sp. vettoriale normato

$$\|f\|_p := \left(\int_E |f|^p \right)^{1/p}$$

$$(\|f\|_{L^p(E)})$$

$$(\|f\|_{p,E})$$

norma su $L^p(E)$

→ purché $p \geq 1$

→ purché

$f \neq g \Leftrightarrow f \neq g$ q.o. su E .

Teorema: $(L^p(E), \|\cdot\|_p)$ è un Banach.

Appartenenza a $L^p(E)$

$$f \in L^p(E) \iff \int_E |f|^p < +\infty$$

Convergenza in $L^p(E)$

$$f_n \in L^p(E), \quad f \in L^p(E)$$

$$f_n \longrightarrow f \text{ in } L^p(E) \quad \left(\lim_n f_n = f \text{ in } L^p(E) \right)$$

$\Uparrow$ per def.

$$\lim_n \|f_n - f\|_p = 0$$

ovvero

$$\left| \lim_n \int_E |f_n - f|^p = 0. \right|$$